Do Experts Govern Better? Medical Mayors and Public Health in Brazil*

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August 28, 2026

Abstract

Do professional experts govern more effectively when elected to political office? We examine this question by studying the election of medical professionals as mayors across Brazil's municipal governments between 2000 and 2022, where nearly 10% of mayors have backgrounds in healthcare. We argue that expert professionals have the ability to govern more effectively in their domain through *sector-specific competence*: a combination of technical knowledge, issue priority, and professional capital. Using administrative, electoral, and public health data from more than 5,500 municipalities, we implement staggered difference-in-differences to estimate the effects of medical mayors on public policy and health outcomes. To probe mechanisms, we complement these analyses with an original survey of more than 500 healthcare bureaucrats across 114 municipalities and an analysis of over 34,000 mayoral campaign manifestos. We find that medical mayors increase healthcare spending, expand health infrastructure, improve healthcare utilization, reduce low birthweight rates, and commit fewer severe infractions in the health sector. These gains do not extend to unrelated policy domains and primarily surface after sustained exposure – rather than one-term exposure – across multiple mayoral terms. Our findings demonstrate that while expertise can improve public service delivery, its benefits may be constrained to specific domains and unfold over long political time horizons.

Keywords: Comparative political economy, local government, representation, public health, Brazil
Word count: 11,436 (not including references)

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*We are grateful to Marco Alcocer, Elena Barham, Hannah Baron, Jamil Civitarese, Alicia Cooperman, Agnes Cornell, Matthew Fuhrmann, Julian Gerez, Simon Gren, Daniel Hirschel-Burns, Hanna Kleider, Zoila Ponce de León, Andres Uribe and participants at UNC-Chapel Hill's Methods and Design Workshop (2024), the Texas Comparative Politics Circle (2025), the UT-Austin Innovations for Peace and Development Research Colloquium (2025), REPAL (2025), APSA (2025), the Representation, Democracy, the Economic Backgrounds of Politicians Workshop at Vanderbilt University (2025), and the University of Gothenburg's Quality of Government Seminar (2026) for feedback on this paper. We thank Instituto Rio 21, particularly Philippe Guedon, for wonderful work in administering our survey. We are grateful to Flávia Batista and Yuri Saldarriaga for research assistance. This project was deemed exempt by IRBs at the University of Texas at Austin and the University of Maryland.

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Introduction

Does professional expertise translate into better policy outcomes? Politicians vary widely in what they know, their occupational backgrounds, and their expertise before entering politics, and this variation can be consequential. On the one hand, leaders' personal characteristics shape what governments do (Krcmaric, Nelson and Roberts, 2020; Fuhrmann, 2020; Hyytinen et al., 2018; Szakonyi, 2021; Novaes, 2024), and candidates' occupations serve voters as a heuristic for competence (McDermott, 2005; Boas, 2014; Mechtel, 2014; Atkeson and Hamel, 2020; Costa and Pereira, 2024). On the other hand, expertise may be solely leveraged as a campaign tool (Desai, Frey and Tyson, 2026), partisanship or political allegiances may reign (de Benedictis-Kessner and Warshaw, 2016), or institutions may constrain any effect that expertise could have (Gerber and Hopkins, 2011). Whether expertise matters at all, and if so in which domains, remains an open question.

Certain experts may possess skills capable of addressing some of the largest threats to human well-being. As countries across the globe are faced with urgent problems – such as environmental disaster, food insecurity, and health crises – is it possible that experts in these fields could solve them? We investigate this question by focusing on one of the most ever-salient issues: public health. Even absent the COVID-19 pandemic, public health has been a consistent, salient issue across contexts. Particularly in low- and middle-income countries, citizens are often burdened with preventable disease and face difficulties in accessing basic healthcare services, often forgoing them altogether due to issues such as long wait times or inadequate availability of basic resources.[1] In most country contexts, addressing these issues is the government's problem; but some representatives may be more able to solve them than others.[2]

Leveraging the presence of medical professionals in elected office – doctors, nurses, and other specialists with expertise in healthcare – we evaluate if medical experts govern better in their domain compared to their peers, generating potential improvements to some of these stubborn issues. While existing work has typically focused on types of expertise that benefit the few rather than the many (Szakonyi, 2018;

[1] https://www.paho.org/en/topics/essential-public-health-functions

[2] Even in mixed-healthcare systems, over 50% of inpatient and 40% of outpatient services are delivered via the public sector, according to the WHO (with the exception of countries in the Eastern Mediterranean). See https://www.who.int/data/gho/data/themes/topics/indicator-groups/sources-of-care-in-mixed-health-systems.

Hansen, Carnes and Gray, 2019), or have highlighted how expertise may constrain the responsiveness of politicians to their constituents (Pereira and Öhberg, 2024), we evaluate if expertise can meaningfully improve outcomes for the public.

We argue that experts in office should produce better outcomes within their domain through what we call *sector-specific competence*. We theorize this as a bundle of three components. First, *issue priority*: experts care more about their domain and are more likely to make it a focus of their administration (Velimsky et al., 2025). Second, *technical knowledge*: experts understand which interventions are effective, which expenditures are wasteful, and how to evaluate the actions of bureaucrats and contractors in their field (Göhlmann and Vaubel, 2007; Avellaneda, 2012). And third, *professional capital*: experts arrive in office with professional ties to capable practitioners, administrators, and organizations that they can draw on to staff and run programs. Their expertise can also inspire a sense of trust and deference among their peers in their area of expertise (Fong, 2020), facilitating more efficient, and perhaps effective, policymaking. These three components are analytically distinct, and we discuss them separately, but in practice, they travel together.

Our argument, however, is not one about *general* competence. Some politicians may exhibit competence in broader areas – such as management or administration – carrying wide-ranging benefits (Carreri, 2021; Carreri and Payson, 2024); others with high levels of education or prior political experience may govern more effectively, as well (Avellaneda, 2009; Frey and Leal, Forthcoming). But these factors should not predict the kind of domain-specific advantage we describe in the paper. Our argument is also distinct from that of politician effort or prioritization alone. While we posit that sector-specific competence must incorporate issue priority to be realized, we distinguish ourselves from other studies (e.g., Szakonyi, 2021; Kirkland, 2021) by examining how such prioritization in concert with our other theorized components (technical knowledge, professional capital) should translate into downstream outcomes.

We test our argument using a series of complementary approaches across a sweeping set of indicators in the context of Brazil, where medical professionals represent 10% of mayors over the past two decades. Not only is their presence common, but municipal governments hold substantial authority and discretion within one of the world's largest public health systems (SUS) (Fujiwara, 2015; Niedzwiecki, 2018;

Toral, 2024*a*),[3] and outcomes such as healthcare delivery, public health infrastructure, and population health are well measured at the municipal level and respond to local government action (Fujiwara, 2015; Toral, 2024*b*; Frey and Leal, Forthcoming). This setting allows us to study our hypothesized main effect, as well as disentangle expertise from leading alternative explanations. Importantly, experts tend to be more educated, politically experienced, and resourced than the average politician (Carnes and Lupu, 2016; Lahoti and Sahoo, 2020; Bastos and Sánchez, 2024; Chaudhuri et al., 2025). As a result, any effect of expertise can easily be confounded with effects of these potentially meaningful traits. But, our case provides a context in which a specific kind of training is widely held by elected officials, outcomes in and outside their domain can be measured over a long period, and comparable groups of experts in other domains are available for comparison. That said, Brazil's local institutional context is not unusual: decentralized governments span the globe (Schneider, 2003), with locally managed public health infrastructure in contexts as diverse as Canada, Finland, and Chile. And, medical professionals in elected office can be found across diverse country contexts, although their presence remains understudied.[4]

We begin by using detailed administrative data – including sector-specific budgets, infrastructure, public health outcomes, and audit performance – to observationally study the impact of medical mayors across Brazil's territory. Covering over 5,500 municipalities from 2000-2022, we identify medical mayors using both their registered occupations and their ballot names, which candidates frequently use to signal professional identity (e.g., "Doctor Paulo," "Nurse Denice") (Boas, 2014; Novaes, 2024; Desai, Frey and Tyson, 2026). We use a staggered difference-in-differences estimator with dynamic or event-study effects (Callaway and Sant'Anna, 2021) that exploits the timing of medical mayor elections across municipalities. We also build on this approach to distinguish between long-run effects of first exposure vs. sustained exposure, disentangling whether and how effects may persist after a medical mayor leaves office, or if

[3] As we will discuss, they are required by law to spend at least 15% of their budgets on health, appoint the municipal Secretary of Health, and govern the staffing and infrastructure of local clinics.

[4] For example, Chile's previous president, Michelle Bachelet, was a long-time physician. In the U.S.'s 119th Congress, 20 members were medical professionals. Gro Brundtland, Norway's three-time Prime Minister, was also a physician, and went on to be the director-general of the WHO.
Notable exceptions of studies of healthcare professionals in office exist, but typically focus on the U.S. and Western European legislative contexts, with legislative output as the key measured outcome. For example, see Fountaine, Shepherd and Skinner (2025) and Costa and Pereira (2024).

presence in office itself is required to realize any benefits. Via these strategies, we examine if their presence translates into meaningful differences in public health outcomes, as well as comparable outcomes in other sectors; we also compare their performance to that of other highly-educated professionals.

Second, we complement this with an original survey of over 500 healthcare bureaucrats across 114 municipalities, evenly divided between those who work under a medical mayor, or who have not for at least the past ten years. Bureaucrats were asked for their perceptions of their mayor, their municipality's capacity to meet health benchmarks, and other related measures. Finally, we incorporate an analysis of over 30,000 of campaign platforms across three election cycles using both semi-supervised topic modeling (Watanabe and Baturo, 2024) and context-specific word embeddings (Rodriguez, Spirling and Stewart, 2023). Via this analysis, we examine if issue prioritization emerges even prior to election, and if prospective medical mayors exhibit differential ways of discussing the healthcare issue in line with explanations of expertise. This combination of strategies allows us to build a robust descriptive and causal picture of sector-specific competence, highlight each of its components, and rule out competing explanations.

Three main findings emerge. First, in line with our theory of sector-specific competence, medical mayors improve outcomes in their domain. They spend more on healthcare, expand basic health infrastructure, increase healthcare utilization, and reduce low birthweight rates, a robust indicator for healthcare performance and service delivery.[5] They are also less likely to severely mismanage public resources in the health sector. Bureaucrats working under these mayors similarly report better perceptions of service delivery and fewer healthcare resource related problems; they are also more approving, expressing opinions consistent with high levels of technical expertise and professional capital.

Second, these effects do not extend beyond healthcare. Although medical mayors increase spending on other policy areas (education and social assistance), this spending does not translate into measurable performance improvements in those sectors. This finding suggests general competence is not at play. Third, expertise takes time and sustained exposure to translate into outcomes, and the long-term effects of first exposure are limited. Results require the active and sustained presence of medical mayors in office to generate benefits; their legacies often do not survive successors with other professional backgrounds.

[5]https://www.who.int/data/nutrition/nlis/info/low-birth-weight

These findings suggest that institutional change in local governance has a time frame longer than one standard electoral cycle, and that the political viability of expertise-based reform depends on continuity in office. In other words, making changes in public outcomes requires time, and in this case, long-term exposure.

We rule out various alternative explanations. In addition to discounting the role of *general* competence, we also investigate if effects are driven by political experience, rent-seeking behavior, and higher levels of education. First, using a close-election regression discontinuity design, we find that prior political experience does not explain results. Second, we investigate rent-seeking, i.e., whether medical mayors might channel resources to healthcare to benefit themselves or their professional networks (Jochimsen and Thomasius, 2014; Hansen, Carnes and Gray, 2019; Szakonyi, 2021). If this were the case, we might expect that corruption should rise in the health sector; we find the opposite. And finally, we consider the independent role of higher levels of education by examining the election of lawyers – a profession with a similar required level of education as many medical professionals. We find no evidence of improved performance in healthcare spending or community health outcomes.

This paper makes three contributions. First, we offer evidence on a question of broad interest in political economy: whether and when expertise translates into better policy outcomes. Most existing work on the personal characteristics of politicians has focused on identity-based traits such as gender, race, or class (Preuhs, 2006; Carnes and Lupu, 2015; Funk and Philips, 2019; Barnes, Kerevel and Saxton, 2023), or has examined occupational backgrounds whose effects appear to favor narrow constituencies (Szakonyi, 2018; Fuhrmann, 2020; Novaes, 2024). We show that expertise can produce broad-based improvements in public service delivery, and that the limits of those improvements line up with the limits of the expert's domain. Moreover, much research in this domain focuses on spending as a proxy for overall performance (Kirkland, 2021). We extend this by looking at downstream outcomes and staff perceptions, disentangling issue prioritization as a necessary, but insufficient, component of expertise. Others have focused on legislative output (Costa and Pereira, 2024; Fountaine, Shepherd and Skinner, 2025); we push this further by examining outcomes among those intended to be affected by such policy.

Second, our concept of *sector-specific competence* sharpens what scholars have often treated as a generic expertise or competence channel (Göhlmann and Vaubel, 2007; Avellaneda, 2012; Carreri, 2021) and clarifies why expertise should produce within-domain rather than across-domain gains. This offers important insights into debates disentangling what constitutes politician quality (Auerbach, Singh and Thachil, 2025; Chaudhuri et al., 2025). It also builds on work isolating under what circumstances politicians highlight their competence (Desai, Frey and Tyson, 2026) by investigating the impacts of these signals. Third, we provide evidence that expertise-based reform operates on a time horizon longer than the electoral cycle. This connects to recent work on bureaucratic continuity and patronage (Toral, 2024*a*,*b*) and suggests that political institutions designed to evaluate politicians such as reelection, term limits, and audit-based accountability may be poorly calibrated to the kind of slow-moving reforms that expertise enables. The point extends beyond medical mayors. Any expertise-based intervention judged over a single electoral cycle risks being declared a failure simply because not enough time has passed.

Theoretical Framework

What makes a politician effective? Scholars have long pointed to a variety of descriptive characteristics – both relatively immutable traits (such as gender and race) and more mutable ones (such as class, education, and occupation). This research suggests differential patterns in outcomes such as spending, policy priorities, and legislation sponsorship (Preuhs, 2006; Carnes and Lupu, 2015, 2016; Funk and Philips, 2019; Fuhrmann and Kim, 2026). For example, female mayors have been shown to spend more on women's issues compared to their male counterparts (Funk and Philips, 2019) and pass more "women-friendly" policies (Meier and Funk, 2017). On the other hand, other characteristics – such as higher education – have more mixed effects (Carnes and Lupu, 2016).

One potentially important, yet difficult to evaluate, characteristic is expertise. Indeed, politicians are often elected to office having had previous experience in specialized areas, such as law (Hain and Piereson, 1975), business (Szakonyi, 2018; Fuhrmann, 2020), and education (Chambers-Ju, 2024). One common – but underexplored – area is medicine.[6] This area is particularly salient, as not only do many governments

[6] For an exception in the U.S. Congress, see Fountaine, Shepherd and Skinner (2025).

worldwide have significant responsibility over the healthcare sector, but the representation of medical professionals is sizable and widespread. For example, in the U.S. 119th Congress, 20 elected representatives are physicians; Chile's previous president, Michelle Bachelet, was a long-time practicing physician; and as previously mentioned, in Brazil 10% of mayors are medical professionals.

While prior research shows that leaders' individual characteristics can shape both their behavior in office and broader policy outcomes, we know far less about the independent role of expertise. One reason is that expertise is difficult to isolate empirically, as it is often closely intertwined with other attributes such as educational attainment, socioeconomic status, or political experience. We seek to disentangle this effect, and ask: can expertise itself in a specific policy domain — such as health — meaningfully shape policy outcomes within that same domain?

We contend that professional expertise can translate into what we call *sector-specific competence*, or competence related to specific skills due to professional training and experience that uniquely equip an individual to make targeted impacts in relevant policy areas. This is distinct from a preference to advance the goals of an industry in question (e.g., Szakonyi, 2018), and is rather the specific ability to execute on a policy goal related to one's professional skill set. Existing research suggests that competence in particular skill sets can drive public-sector performance (Avellaneda, 2009, 2012; Göhlmann and Vaubel, 2007), but stop short of explicitly positing the mechanisms behind this relationship. We theorize that sector-specific competence has three main components: technical knowledge, issue priority, and professional capital.

First, we posit that sector-specific competence can be partially attributed to *technical knowledge*, or the insights, skills, and expertise related to a field that are imparted through both professional training and on-the-job experience. This knowledge translates into a deeper understanding of the practical considerations involved in designing and implementing policy, including which types of policies or interventions are most likely to succeed, which needs within a sector are most pressing, which investments are likely to be inefficient or unnecessary, and how to effectively evaluate the performance of bureaucrats, contractors, staff, and infrastructure. For example, in the context of the U.S. and Europe, Göhlmann and Vaubel (2007) find that central bank council members with former experience as central bank staff tend to generate lower inflation rates than those with other backgrounds. This is attributed to their on-the-job technical knowledge. Similarly, Jochimsen and Thomasius (2014) find that finance ministers with previous financial expertise are able to lower deficits more effectively. However, these studies typically employ a correlational approach, limiting their ability to truly parse out the independent role of technical knowledge.

Second, sector-specific competence is also characterized by *issue priority*. To exhibit a heightened level of competence in an area, a politician must dedicate resources (e.g., time, money) to that area to make a difference. This often necessitates dedicating less attention to other areas. In this vein, research on physicians in the U.S. Congress has shown that they are more likely to author healthcare-related bills (Fountaine, Shepherd and Skinner, 2025). And, recent work in Europe echoes this finding, showing that a politician's previous occupation affects their issue attention, leading to increased emphasis on policy issues related to their previous profession (Velimsky et al., 2025). The prominence of issue priority may be attributed to two complementary reasons. First, a previous career in a field can strongly socialize individuals to personally believe in its importance, leading them to consciously or subconsciously prioritize the issue area over others (Peterson, 1992; Kitschelt and Rehm, 2014; O'Grady, 2019; Barnes, Kerevel and Saxton, 2023; Fountaine, Shepherd and Skinner, 2025). Socialization within a career can "construct identities that dramatically affect values and attitudes" (Gade and Wilkins, 2012), and at times be a stronger predictor than attitudes attributable to social origins (Meier and Nigro, 1976). Second, politicians of certain recognizable careers, such as doctors, may be elected to voters in part due to their expertise. Thus, prioritizing and delivering results in this area is also important for these elected representatives to communicate their value to their voters (Boas, 2014; Barnes and Saxton, 2019; Turner, 2024).

Finally, politicians with sector-specific competence may also benefit from *professional capital* that facilitates their effectiveness in an area. Professional capital can provide not only access to valuable networks and institutional knowledge (Novaes, 2024; Szakonyi, 2018), but also credibility and standing among relevant actors, generating greater deference from peers and stakeholders (Fong, 2020; Makse, 2022). Just as training and on-the-job experience facilitates technical knowledge, it also allows such professionals to develop connections (e.g., with classmates, colleagues) and have prolonged contact with the sector's infrastructure and relevant actors. This can provide a form of "informal" embedded capacity, which can have positive impacts on policy outcomes, including healthcare (Kuo and Kelly, 2026). Further, professional experience can generate the perception of effectiveness among relevant policymaking peers and functionaries, generating greater deference, willingness to collaborate, and more compliance with priorities on the issue area in question. Indeed, Costa and Pereira (2024) show that legislators with specific sets of occupational expertise enjoy greater support from their peers, with politicians more likely to co-sign legislation proposed by peers with relevant professional expertise. Much of this work has relied on legislative contexts Fong (2020); Costa and Pereira (2024), however its relevance also likely applies to executive office, as well, where policymakers must still consistently work with other relevant stakeholders, including cabinet members, members of the legislature, executives at other levels of government, and their own functionaries.

In all, we consider that these three elements – *technical knowledge*, *issue priority*, and *professional capital* – constitute the three core pillars of *sector-specific competence*. In all, they should lead a politician with a previous professional background to be particularly strong performers in related policy areas. Applying this to consider medical professionals elected in office, we posit that medical professionals should outperform other mayors in core public health outcomes. Specifically, we contend that not only should medical mayors invest more in public health – both in terms of financial and physical resources (e.g., personnel) – but that their investments and competence should translate into measurable public health outcomes. These should include outcomes that existing work has shown are susceptible to the influence of public policy (e.g., Fujiwara, 2015), such as healthcare utilization (e.g., prenatal visits), prenatal outcomes (e.g., birthweight), and disease (e.g., number of cases of preventable diseases).

What Sector-Specific Competence is Not

While we argue that sector-specific competence can meaningfully shape politician performance, it is important to distinguish this concept from several adjacent characteristics that have also been linked to politician quality and effectiveness.

General Competence: While sector-specific competence may drive strong performance in an expert's domain, it should not translate directly to benefits outside of that area. In this way, it is distinct from *general competence*. A politician with sector-specific competence should not be – on average – more competent across other issue areas than their peers. Their education or experience does not provide them with an advantage in other policy areas. In fact, an observable implication of sector-specific competence should be a lack of systematic returns in areas outside of the sector of the politician in question. While perhaps they may value other policy areas as well, they should be no more likely to deliver positive returns to non-health areas on average.

Education: While sector-specific competence in certain industries may entail higher levels of educational attainment, outcomes should be independent of any effect of education. Sector-specific competence should apply equally to experts in industries that may not require high levels of education, such as laborers. Indeed, the case of healthcare professionals that we evaluate includes occupations with varying educational requirements (e.g., public health worker, technician, doctor). Further, research on the role of education alone has debated its importance. While some have demonstrated that politicians with higher educational attainment are more efficacious (Lahoti and Sahoo, 2020; Sørensen, 2023), others have pushed back on the importance of education in explaining politician's efficaciousness (Carnes and Lupu, 2016). If sector-specific competence drives performance in a specific policy domain, we should not expect the same gains in relevant actors from highly-educated professionals, writ large.

Political Experience: Some careers are known for particularly high levels of political involvement (e.g., lawyers), which may make individuals in these categories more effective not due to their competence in their sector, but rather due to political competence. For example, Olvera and Avellaneda (2019) find that politicians with previous government experience are better performers across a variety of indicators, including school enrollment and infant mortality levels.[7] This could very well apply to the case of medical professionals, given their prominence in elected office across the globe. For sector-specific competence to be at play, its impacts should not be explained by previous tenure in elected office.

Rent-Seeking Behavior: Finally, sector-specific competence should be independent from a politician's efforts to benefit an industry that they represent (Jochimsen and Thomasius, 2014; Hansen, Carnes and Gray, 2019; Krcmaric, Nelson and Roberts, 2020). Rather than simply the allocation of resources at

[7] See also Frey and Leal (Forthcoming).

the expense of the success of other policy areas or groups of citizens – which politicians in areas such as business have been shown to do (Szakonyi, 2021, 2018; Novaes, 2024) – sector-specific competence should not be systematically characterized by malfeasance, including selectively benefiting certain contractors or private actors. As we argue, too, investments should translate into impact for sector-specific competence to be at play.

Setting: Medical Mayors & Local Public Health in Brazil

Brazil provides an ideal scenario to study the role of expertise in office, and provides particularly well-suited conditions to study this in the domain of healthcare. Specifically, we focus on mayors – the executive authorities over Brazil's more than 5,570 municipalities. These officials are elected via majoritarian elections every four years. Between 2000-2022, about 23% of municipalities had at least one medical candidate running for mayor. Further, they comprise about 10% of elected mayors, with 18% of them being re-elected for a second term.[8] These mayors are detectable via information collected during candidate registration. During this process, candidates register both their occupation and the name they wish to use on the ballot, or their *nome de urna*, with the Superior Electoral Tribunal (TSE). Candidates are given discretion to choose their ballot names, and often include occupational identifiers in order to communicate to voters their professional experience. For example, candidates within the medical field may register their ballot name with the title "Doctor" or "Nurse." We use these identifiers to isolate all candidates with medically relevant occupations and ballot names.[9]

Figure 1 illustrates the distribution of elected medical mayors across the country between the 2000 and 2020 elections, as well as the specific occupations they registered with the electoral court during this time period.[10] As seen in Table 1, medical mayors originate from across the ideological spectrum, with 18% of them representing the center MDB party (Brazilian Democratic Party, *Movimento Democrático*

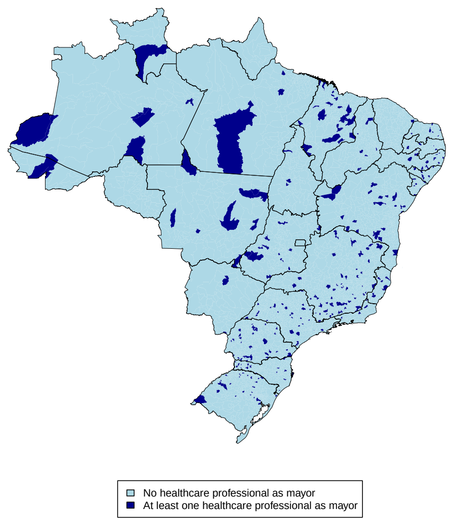
[8] Between 2004 and 2020, re-election rates for medical mayors varied from 28 percent (2004) to 5 percent (2012).

[9] See Appendix A.1 for the full methodology

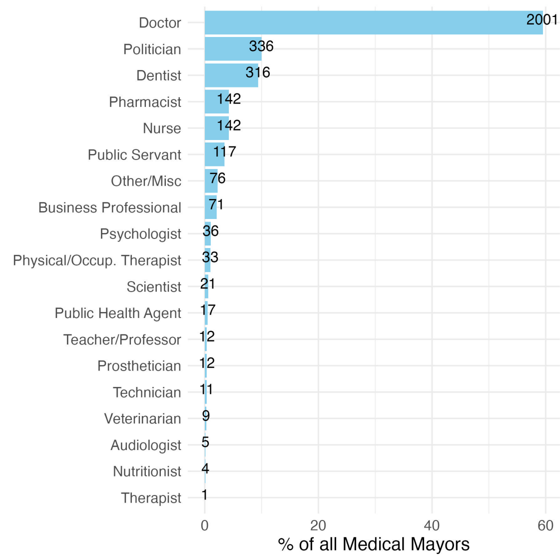
[10] As will be discussed in the methods section, medical mayors are also identified by their "ballot names" allowing them to register an occupation – such as politician – that may be different from a medical profession, while still signaling their medical background.

Brasileiro). Appendix A.2 provides evidence that medical mayors are evenly distributed across municipalities in terms of population, area, human development, and educational outcomes.

Figure 1: Medical Mayors by Geography and Occupation



(a) Municipalities that elected a healthcare professional as mayor at least once 2000-2020



(b) Medical mayors by registered occupational category

Table 1: Top 15 Parties of Elected Medical Mayors (2000-2020 Elections)

Party	L-R Placement	N. Medical Mayors	% of Medical Mayors
MDB	Center	2186	18.13
PSDB	Center-Right	1914	15.87
DEM	Right	1084	8.99
PP	Right	1008	8.36
PSB	Left	862	7.15
PT	Left	758	6.29
PTB	Center-Right	674	5.59
PDT	Center-Left	644	5.34
PSD	Center-Right	622	5.16
REP	Right	518	4.30
CID	Center	434	3.60
PL	Right	396	3.28
PV	Left	216	1.79
PSC	Right	136	1.13
PSL	Right	124	1.03

Note: Author's own calculations utilizing data from the TSE. Medical professionals are identified via their ballot names and registered occupations, as detailed in the "Electoral Data" section of the paper. Ideological placements are based on Zucco and Power (2024) and contextual knowledge of parties not included in their estimates.

Moreover, not only are medical professionals common and widespread municipal executives, but they also hold significant sway over healthcare policy. Brazil's universal public healthcare system – the Unified Health System (*Sistema Único de Saude*, SUS) – is significantly decentralized, with major responsibilities held by municipal authorities (Avelino, Barberia and Biderman, 2014; Niedzwiecki, 2014). About 75% of the population rely solely on this system for healthcare,[11] and 34% of Brazil's municipalities do not even have private medical providers.[12] Municipalities are obligated to guarantee basic healthcare services for their communities and must spend at minimum 15% of their annual budget on public health, although on average they spend over 20%.[13] They are also required to have health departments, tasked with managing municipal healthcare facilities and are responsible for providing primary health care, health surveillance, and guaranteeing patient access to hospital care.

Beyond the municipality having significant purview over this policy area, the ways in which mayors specifically can affect it are multiple. Mayors have significant input on the governance and staffing of municipal health infrastructure. For instance, they are tasked with appointing the municipal Secretary of Public Health, a post outlined in the constitution as a key player in administering SUS.[14] Beyond the secretary, mayors have significant capacity to appoint other key bureaucrats across the public sector; indeed, political connectedness among bureaucrats to the mayor in specific sectors, such as education, has been shown to increase bureaucratic effectiveness (Toral, 2024*a*). They also have the power to determine how many civil service hires they will make in this sector.[15] And, these efforts are not without consequence. Municipal health reforms have demonstrable impacts on both healthcare utilization and individual-level health outcomes. For example, Fujiwara (2015) showed that increased enfranchisement of lower-educated voters led to changes in the responsiveness of municipal politicians to their population, changes in fund-

[11]https://www.commonwealthfund.org/international-health-policy-center/countries/brazil

[12]https://www.gov.br/saude/pt-br/assuntos/noticias/2023/marco/brasil-possui-1-915-municipios-sem-servicos-medicos-privados-que-dependem-exclusivamente-do-sus

[13]https://www.institutomaissaude.org.br/voce-sabe-como-funciona-a-saude-publica-municipal/ https://www.commonwealthfund.org/international-health-policy-center/countries/brazil

[14]https://www.gov.br/saude/pt-br/sus

[15]They are also cognizant of this power. Toral (2024*a*), for example, finds that 76% of surveyed mayors indicated they had the highest level of responsibility of improving public service delivery quality.

ing patterns, and ultimately increased both healthcare utilization and improved newborn health among lower-educated populations. Thus, we should expect that mayors' efforts to improve healthcare service delivery should be able to have similar downstream effects.

Important to our study, municipal governance in Brazil also provides us with ample opportunity to test our competing explanations. Mayors also have purview over other areas, such as social policy and education. For example, municipalities play a significant role in the administration of the *Bolsa Familia* program, particularly helping to enroll beneficiaries.[16] They also oversee municipal public schools. Further, medical professionals are not the only experts commonly elected to office; lawyers, for example, comprise 4 percent of elected mayors. In all, Brazil provides a context that not only allows us to study medical professionals' performance in healthcare specifically, but to compare their performance across fields, and across other types of experts.

Empirical Evidence

To test our theoretical expectations, we draw on three types of data and analyses. First, drawing on rich, time-series administrative data, we leverage a staggered difference-in-differences design to evaluate the impact of medical mayors on a variety of both institutional and community health outcomes over time. Second, we field an original survey of local level public health bureaucrats across over 100 municipalities. And third, we collect and analyze a large corpus of campaign platforms of both medical and non-medical mayoral candidates across three elections. Table 2 maps the evidence we leverage to each core claim.

Time-Series Administrative Analysis

Data: To evaluate the impact of medical mayors, we collect detailed administrative data regarding the presence of healthcare professionals running for mayoral office and various municipal indicators related to public health and other sectors. Much of our data spans over two decades (2001-2022), although some

[16]While *Bolsa Família* is a federal conditional cash transfer program, municipalities have a significant role (Fenwick, 2009), particularly in helping to enroll beneficiaries. For more information, see here: Confederação Nacional de Municípios link

Table 2: Mapping Evidence to Claims

Claim	Test	Data	Evidence
Sector-Specific Competence			
Medical mayors impact community and institutional health outcomes	DID; Correlation	Administrative data; bureaucratic survey	Figs 2-3, Fig 7
...but not in other policy areas	DID	Administrative data	Fig 4
Components and Mechanisms			
Issue priority	Topic models	Campaign platforms	Fig 5
Technical knowledge	Correlation; Word embeddings	Bureaucratic survey; Campaign platforms	Figs 7 & 6
Professional capital	Correlation	Bureaucratic survey	Fig 7

variables of interest are only available for later years (e.g., 2008-2022). In all, the majority of our analysis covers six mayoral terms, with some covering three or four.

To identify medical mayors, we rely on both the ballot names that candidates use (*nomes de urna*) and occupations that candidates register with the TSE as previously discussed.[17] In all, we identify 9,023 medical candidates (or 7.4% of candidates) across six mayoral election cycles (2000-2020) who fit at least one of these criteria. Of these, 3,362 were elected, representing 10% of elected mayors.

We examine the influence of medical professionals in public office on a series of public health indicators. First, we examine their institutional actions, including spending, the amount of public sector health units in their communities, and the number of individuals employed in the public health sector. In terms of budgets, data on committed municipal budgets is most exhaustive, and is drawn from the Ministry of

[17] In terms of ballot names, we include those who run on a name which includes the term "doctor" (*doutor* or *médico*), "nurse," "pharmacy," "psychologist," "physiotherapist," "dentist," "from the hospital," or "of health" including any related abbreviations and female/male versions of these words. Importantly, we take care to eliminate those who might use the term "doctor" but are not healthcare professionals. For example, we eliminate those who use the title "doctor" (or Dr., Dra., doutora) who simultaneously register their occupations as lawyers or civil police officers, two professions that also use this title. In terms of occupation, we include any candidate who registered their occupation as a medical doctor (médico), nurse (enfermeiro), public health worker (agente de saude e sanitarista), nurse technician (técnico de enfermagem e assemelhados), laboratory and X-ray technician (técnico de laboratório e raios x), psychologist (psicólogo), biomedical professional (biomédico or biólogoco e biomédico), physical or occupational therapist (fisioterapeuta e terapeuta ocupacional or terapeuta), audiologist (fonoaudiólogo), pharmacist (famacêutico), nutricionist (nutricionista e assemelhados or enfermeiro e nutricionista), paramedic (paramédico), dental prosthetics professional (protético), medical and dental equipment operator (operador de equipamento médico e odontológico), and dentist (odontólogo).

Finance.[18] We also collect data on the number of family health centers present in each municipality from the National Registry of Health Establishments (CNES).[19] Also from CNES, we gather the total number of employees in the municipal healthcare system for each year from 2007-2019. Similar indicators – namely data regarding hospital employment – has been shown to be affected by mayoral decision-making in Brazil (Toral, 2024*b*).[20]

Second, we examine public health outcomes to determine the effect they may have on the health of their constituents. We collect information on birthweight and pre-natal visits from Brazil's Live Birth Information System (SINASC). From these data, we calculate the percentage of births considered of low weight (below 2500g), the average birthweight, and the total number of pre-natal visits in a municipality. These measures are common indicators of infant health and healthcare utilization used in public health research. Social science research has similarly shown relationships between political factors, such as voting technology, and these outcomes (Fujiwara, 2015).[21]

Finally, we consider their performance and malfeasance drawing on data from random municipal audits. From 2003 to 2014, the Comptroller General (CGU, Controladoria-Geral da União) randomly selected municipalities for comprehensive audits of federal fund expenditures, conditional on municipality size (Ferraz and Finan, 2011). We focus on severe infractions, which correspond to fraud, illegal or illegitimate use of funds, misappropriation, or embezzlement of public resources. These are the most direct behavioral proxies for rent-seeking.[22] We estimate the effect of electing a medical mayor on the rate of severe infractions per audited item across three sectors: health, education, and social assistance.

Research Design: To assess the effect of electing a healthcare professional as mayor, we use a dynamic

[18] We also collect information regarding funds spent – rather than committed – which is available for a smaller set of years, and collected from the National Treasury's public sector financial database (SICONFI). Accounting and Fiscal Information System for the Brazilian Public Sector, *Sistema de Informações Contábeis e Fiscais para o Setor Público Brasileiro*.

[19] Available beginning in 2007

[20] In the appendix, we include the number of basic healthcare clinics as an extra outcome.

[21] In the appendix, we include the number of tuberculosis cases as an extra outcome.

[22] We rely on audits conducted between 2003 and 2014, the period in which the CGU used a lottery-based selection procedure. Beginning in 2015, the selection became risk-based and oriented toward detecting corruption, which would bias our comparisons.

(event-study) difference-in-differences (DID). This framework allows us to identify the effects of electing a medical mayor on outcomes of interest based on the length of exposure by examining dynamic effects over time. This design is based on comparing changes in outcomes between municipalities with and without a healthcare mayor over time. It relies on a key assumption, the existence of parallel trends, meaning that the treatment and the control group would have followed the same trajectory over time in the absence of the treatment.

Given the structure of our dataset, we have variation in treatment timing across units, which means that municipalities can elect a healthcare professional as mayor in different election years. We use the Callaway and Sant'Anna (2021) difference-in-differences (DiD) estimator, which is designed to handle situations of staggered treatment, and overcomes bias produced by strategies such as the TWFE estimator (De Chaisemartin and d'Haultfoeuille, 2020). This estimator follows a multi-step process. First, it estimates group-time treatment effects for each group and time period. Second, the group-time treatment effects are aggregated to produce overall average treatment effects. In particular, we aggregate them based on the length of exposure and provide dynamic or event-study effects. In other words, this aggregation gives us the average treatment effect at different years relative to the first year of having a healthcare professional as mayor.

In this staggered DiD, we use never-treated municipalities as the control group. Treatment begins in the first mayoral term in which a municipality elects a healthcare professional. Medical mayors are not always succeeded by other medical mayors, so we define exposure in two ways: *sustained exposure* and *any first exposure* (we expand on this below).

Because our outcomes are constructed as *average values over a mayoral term*, the unit of analysis is the municipality-term. This aggregation aligns the measurement of public health outcomes with the time horizon over which municipal policy choices and program implementation are likely to operate. We focus on this unit of analysis as many of the outcomes we study, such as changes in service provision, primary care capacity, and downstream health indicators, evolve gradually and may not respond fully within a single calendar year.[23] We report municipality-year (annual) event-study estimates in Appendix

[23]Year-to-year variation can also reflect short-run noise (e.g., transitory shock or timing of program rollouts)

B.8, which trace year-by-year dynamics within and across mayoral terms.

Formally, let G_i denote the first mayoral term in which municipality i elects a medical mayor, with $G_i = \infty$ for municipalities that never do (i.e., never-treated). The building block of the approach is the group-time average treatment effect on the treated:

$$ATT(g, \tau) = \mathbb{E}[Y_{i\tau}(g) - Y_{i\tau}(\infty) \mid G_i = g],$$

which captures the average effect in term τ for municipalities first treated in term g, relative to the never-treated group. Outcomes are averaged over each four-year mayoral term. We estimate each $ATT(g, \tau)$ with a doubly robust approach that conditions on pre-treatment covariates: municipal health spending, population, and area. Standard errors come from a bootstrap clustered at the municipality level.

We then aggregate the group-time effects into dynamic (event-study) coefficients by length of exposure, $e = \tau - g$, which give the average effect e terms after a municipality first elects a medical mayor. Estimates for $e < 0$ provide evidence consistent with the parallel trends assumption (i.e., null effects for pretreatment periods are consistent with a common trajectory). The overall ATT reported in the figures averages the dynamic coefficients across post-treatment terms.

As a reminder, the Callaway and Sant'Anna estimator assumes staggered adoption: once a unit is treated, it stays treated; however, medical leadership does not always persist. Many municipalities elect a non-medical mayor in the term after a medical one, so we define exposure in two ways. Our main specification measures *sustained exposure*, meaning that a municipality is treated when a medical mayors hold office consequently, through re-election or the election of another healthcare professional. We drop the municipality-terms in which a non-medical mayor governs after a medical mayor, so a municipality-term will exit the estimation sample when medical leadership ends. This is the standard staggered design: treatment status coincides with a medical mayor in office, and the dynamic coefficient at e is the effect of e consecutive terms of medical leadership.

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and, as a result, obscure the trends and long-term patterns. Therefore, by averaging outcomes over the mayoral period, we emphasize medium-run changes that better match the scope of mayoral decision-making and the lagged nature of health improvements.

Our second specification measures *any first exposure*. Here the treatment is the first election of a medical mayor, and a municipality remains in the treated group in every subsequent term regardless of who succeeds them. Its estimand is the average effect of ever electing a medical mayor, pooled over trajectories in which medical leadership continues and trajectories in which it ends. In other words, *sustained exposure* isolates the effect of a medical mayor in office, and *any first exposure* evaluates whether a single election leaves a legacy that outlasts the mayor. We report both in Figure 2; all other estimates use sustained exposure, and Appendix B.4 reports full results for any first exposure. In both specifications, we exclude municipalities that elected a medical mayor in the first term of the panel, as these are always treated and provide no pre-treatment information.[24] Appendix Figure B1 illustrates the resulting treatment structure.

While many studies primarily rely on close-election regression discontinuity designs (RDDs) to assess the impacts of certain types of candidates, we rely on the staggered DID for a handful of reasons. First, this approach allows us to assess dynamics over time in a disciplined fashion. This is relevant as our outcomes of interest – particularly slow-moving indicators, such as birth weight – may not be realized in short time frames. It also allows us to capture multiple terms of treatment (e.g., the succession of multiple medical professionals as mayor), and to effectively distinguish between *sustained exposure* and *any first exposure*. Furthermore, it provides conditions to make broader conclusions beyond the bandwidth that an RDD allows, challenging its external validity (De la Cuesta and Imai, 2016). While an RDD may account for issues regarding selection into treatment, we demonstrate that our sample is balanced across a series of pre-treatment characteristics which may induce selection-bias, including political factors, population, human development, and educational attainment (Table A1). Indeed, as can be seen in Figure 1, treated municipalities display a widespread presence across the country. We also provide a covariate balance plot comparing municipalities with and without a medical mayor across key pretreatment covariates (Figure A3). The standardized differences are below 0.2, the conventional threshold for evidence of covariate balance (Silber et al., 2013). Of course, this only speaks to observed characteristics, but it provides suggestive evidence that municipalities with medical mayors are not larger or more populated, did not have higher

[24]We also remove cases of special or irregular elections.

prior health spending, do not have higher development indices or education levels, and did not exhibit different voting patterns for the main political parties.

Bureaucratic Survey

We fielded an original bureaucratic survey of 520 municipal-level healthcare bureaucrats across 114 municipalities, covering 21 states.[25] The survey was designed such that about 50% of respondents worked in municipalities currently governed by a medical mayor, while the other 50% worked in municipalities where there were no medical mayors for at least the past 10 years.[26] Via this survey, we compare officials employed in municipalities with sitting medical mayors to those in the latter strata to determine how public health administration in the two differs. Respondents were contacted via a mixed-methods approach, leveraging both online and in-person recruitment, and completed the survey via Qualtrics.[27]

We targeted mid-level bureaucrats – such as employees of municipal health ministries and clinic managers – as they sit at the middle ground of political influence and on-the-ground health administration. Cabinet members and political appointees, for example, may be more likely to report favorable opinions of their mayors due to their close political ties (Toral, 2024*a*). Mid-level bureaucrats, such as street-level managers and ministry employees, have on-the-job-informed opinions of mayoral quality based on a wide variety of factors.[28]

Within the survey, we analyze three main concepts of interest: mayoral approval (four items), reported municipal health service delivery capability (four items), and reported issues with public health resources (six items). We analyze each item separately, and create an overall index for each of these concepts.[29] These

[25]This survey was fielded in collaboration with Instituto Rio21, a leading survey and field research firm based in Rio de Janeiro.

[26]To account for the recent 2024 municipal elections, for the first group, we restrict our samples to municipalities that had recently re-elected a medical mayor.

[27]Initially, respondents were recruited via e-mail, WhatsApp, and LinkedIn. The survey team then expanded to recruit participants and distribute the link via site visits to municipal public health offices, attendance at a municipal-level public health conference, and by contacting municipal-level healthcare professional organizations.

[28]In exchange for a municipality's participation, we sent the Secretariat a brief report from the survey, including a summary of aggregated results of select survey questions.

[29]For approval, we analyze four questions on a 1-5 agreement likert scale: "The mayor works to improve health-

measures help us isolate both subjective assessments of the mayors directly, but also general perceptions about the municipality's ability to meet basic healthcare priorities. To isolate whether there are robust correlations between a mayor's medical profession and these outcomes, we regress respondent answers on an indicator variable for the mayor's occupation, along with a series of controls:

$$Y_i = \alpha + \beta M_i + \mathbf{X}_i'\gamma + \delta_s + \varepsilon_i, \quad [w_i], \tag{1}$$

where Y_i is the standardized response provided by the healthcare bureaucrat (or an index measure created from their responses), M_i is an indicator variable for whether or not their mayor is a medical professional, and $\mathbf{X}_i$ is a vector of control variables.[30] We cluster standard errors at the municipality level. Given that state-level governments also play a role in healthcare service delivery, we include state-level fixed effects (δ_s). To account for imbalance in respondent and municipal characteristics across our two strata, we generate and incorporate survey weights (w_i) based on a series of variables we collect.[31]

care for all of the population," "The mayor has special knowledge to improve the quality of healthcare in this municipality," "I have confidence in the mayor," and "The mayor of this municipality worries/concentrates more on public health than other mayors in neighboring municipalities." For service delivery, we analyze four questions on a 1-5 likert scale rating the municipality's capacity across different areas: "To buy necessary equipment of basic health units/clinics," "To employ the adequate number of healthcare professionals to meet basic patient needs," "To offer necessary pre-natal care," and "to meet demand for vaccines." For reported issues with resources, we ask respondents to select all items in a list that their municipality struggles with. Focusing on public health resources, we analyze both the reporting of each problem and the total number of problems reported: lack of doctors and other related medical professionals, long wait times for medical consultations, lack of clinic/hospital beds, long wait for emergency medical visits, insufficient training for doctors, poor distribution of doctors.

[30]Specifically, we control for the mayor's ideology (based on their party affiliation), whether or not the respondent is a manager-level employee, if the respondent is a political appointee, and the type of healthcare organization they work in.

[31]These are respondent gender, respondent age, respondent race, respondent religious identification, respondent political ideology, municipality region, municipality GDP per capita (2021), and the number of *Bolsa Familia* recipients per 100k residents in the municipality (2024). See Appendix F for details.

Campaign Platforms

We collect official campaign platforms from Brazil's TSE for the universe of available mayoral candidates across three election cycles (2012-2020, across all municipalities), constituting over 34,000 documents.[32] We transformed the text data into an analyzable format via OCR, and merged all platforms with corresponding candidate meta-data. We analyze these documents utilizing multiple text analysis methods to investigate core mechanisms of our theory.

To evaluate *issue priority*, we rely on semi-supervised topic modeling, specifically seeded Latent Direchlet Allocation (LDA) (Watanabe and Baturo, 2024), to estimate the proportion of each platform dedicated to core issue areas, including healthcare. We descriptively model candidates' emphasis on each issue area based on a series of pre-election covariates, incorporating municipality and election-year fixed effects, and municipality-election year clustered standard errors. With respect to *technical knowledge*, we should expect that political candidates with medical backgrounds should, on average, discuss topics related to healthcare in different ways, perhaps emphasizing more technical or specific aspects of healthcare. To explore this, we examine the semantic framing of healthcare in their platforms, drawing on context-specific word embeddings designed to understand the way in which politicians of different groups use different words to discuss the same issue (Rodriguez and Spirling, 2022). To do so, we analyze how candidates discuss the term "health" (*saúde*) via the a lá carte on text (conText) modeling approach (Rodriguez, Spirling and Stewart, 2023). Given our large and rather distinctive corpus of text, we estimate our own set of GloVe embeddings (Pennington, Socher and Manning, 2014) rather than using pre-trained options (Rodriguez and Spirling, 2022).[33] See Appendix G for further conText modeling details.

[32]Beginning in 2009, all candidates running for mayor submit individual governance platforms (*propostas/programas de governo*) to the TSE outlining their policy priorities and platforms.

[33]In all analyses, our text is pre-processed to remove stop-words, state names and acronyms, political party names, and boilerplate language (e.g., "election", "municipality"). Throughout this process, a comprehensive series of terms related to healthcare and other essential policy areas (e.g., education) are retained. We lemmatize words prior to analysis using a Universal Dependencies model (udPipe) for the Portuguese language. Following lemmatization, words are regularized to remove special characters. For `context` models, words are further stemmed.

Impacts on Public Health Outcomes

We begin by presenting the overall effect of medical mayors on institutional and community health outcomes. We do so via the overall average treatment effect on the treated (ATT) using the Callaway and Sant'Anna estimator for staggered difference-in-differences (DID). For all analyses, outcome variables were standardized to facilitate interpretation and ensure comparability across models.[34] Figure 2 presents the ATT across six outcomes capturing different dimensions of local public health performance by mayoral term, in addition to the 90% and 95% confidence intervals. Each panel reports the two estimates defined in the research design: *sustained exposure*, our main specification, and *any first exposure*.

Figure 2: Average Post-Treatment Effects across All Medical Mayoral Terms

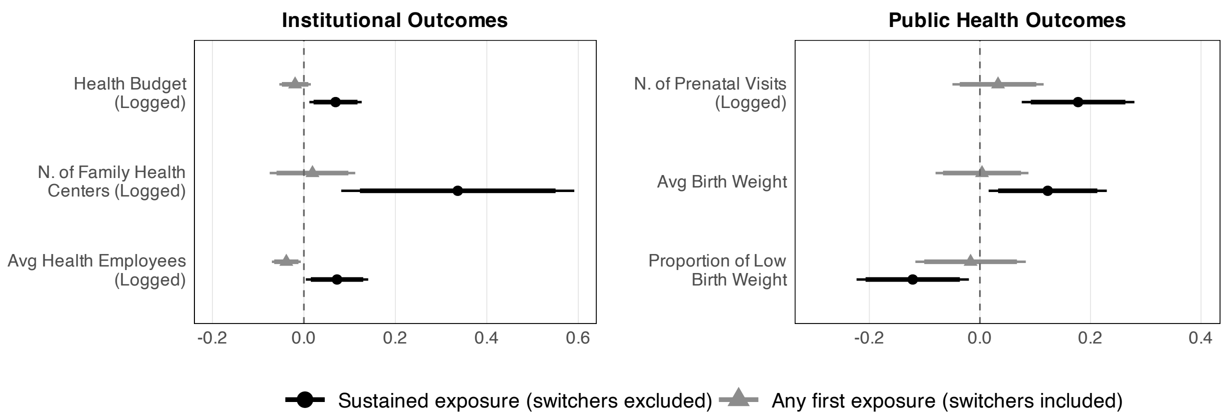


Figure Note: Average post-treatment effects from staggered difference-in-difference estimation. Includes 90% and 95% confidence intervals. Sustained exposure estimates exclude municipalities that revert to a non-medical mayor; any first exposure estimates retain these switchers and average over sustained and non-sustained trajectories. The dataset includes 19,424 observations (municipality–mayoral terms)

Overall, results show that *sustained exposure* to a healthcare professional as mayor has a positive and significant average effect over time. Medical mayors spend more on healthcare, increase the number of family health centers, and the average number of healthcare employees ($p < 0.05$).[35] Moreover, in terms

[34] We applied logarithmic transformations to all variables representing funds and counts of units (e.g., personnel or facilities), and all funds are adjusted for inflation. Variables expressed as percentages, such as the proportion of low birth weight, were not transformed. As a robustness check, in Appendix B.3 we present results using the non-logged values, and the main conclusions do not change. We refrain from adjusting outcomes by population because doing so can introduce post-treatment bias, as population levels may themselves be affected by improvements in healthcare. To account for population differences across municipalities, we include pre-treatment population levels as a control variable instead.

[35] Additional analyses suggest the final effect is driven by increased hiring of healthcare providers (e.g., doctors, nurses, public health agents) (see Appendix B.1.1).

of public health outcomes, effects are at times stronger. At the aggregate level, medical mayors are responsible for an increase in healthcare utilization (prenatal visits), an increase in the average birth weight of children, and a decrease in the proportion of low birth weight. These results suggest that medical mayors' spending is not just an artifact of issue priority; rather, their spending meaningfully translates into investment in tangible infrastructure, personnel, and public health outcomes.

However, we find a meaningful difference between effects of *sustained* and *first exposure*. Compared to *sustained exposure*, estimates of *first exposure* alone attenuate toward zero and lose significance (full results in Appendix B.4). This suggests two core takeaways. First, this attenuation is consistent with the hypothesis that the main estimates capture genuine treatment effects tied to the active presence of a medical professional, not artifacts of sample construction. Second, it suggests that a the one-time election of a medical professional in office is insufficient for meaningful public health gains. Rather, it is the active and sustained presence of medical mayors in office that generate tangible benefits, and short bursts of exposure do not have long-term legacies.

In Figure 3, we zoom in on *sustained exposure*, presenting the dynamic effects for the same outcomes from our DID approach. Here, the unit of analysis is still the municipality–mayoral term, but the x-axis represents the number of mayoral terms elapsed since the first election of a healthcare professional. Grey points represent the pre-treatment periods, which serve as evidence supporting the parallel trends assumption. We observe a clear and consistent null finding when estimating the change in trajectories between treated and control groups before sustained exposure to a healthcare professional as mayor.[36] Post-treatment points (black points) show the treatment effects at different mayoral terms of exposure.

Overall, in the post-treatment periods, we see a stable pattern of little to no effects during the first mayoral term, followed by an increase beginning in the second term of exposure. This pattern is consistent across most outcomes. This aligns with the differences we observe between first and sustained exposure, demonstrating that in practical terms, effects become apparent when either a medical mayor is re-elected, or succeeded by another medical mayor. This accounts for about 30% of our treated sample

[36]These points compare changes in outcomes between municipalities that will eventually have healthcare professionals as mayors and those that never had healthcare mayors.

(see Figure B2). It is important to note that none of the individual dynamic effects are statistically significant. However, each period's effect estimate in the dynamic treatment effects approach relies only on data from that specific period, meaning that the sample size is substantially reduced for each data point, affecting the precision of each dynamic estimate (Huntington-Klein, 2021). As a result, we use the overall ATT as the main finding, with the dynamic effects serving as an insight into how the effects evolve over time and as a means of better understanding their trajectories.

Figure 3: Effect of Medical Mayors on Institutional and Public Health Outcomes by Mayoral Term

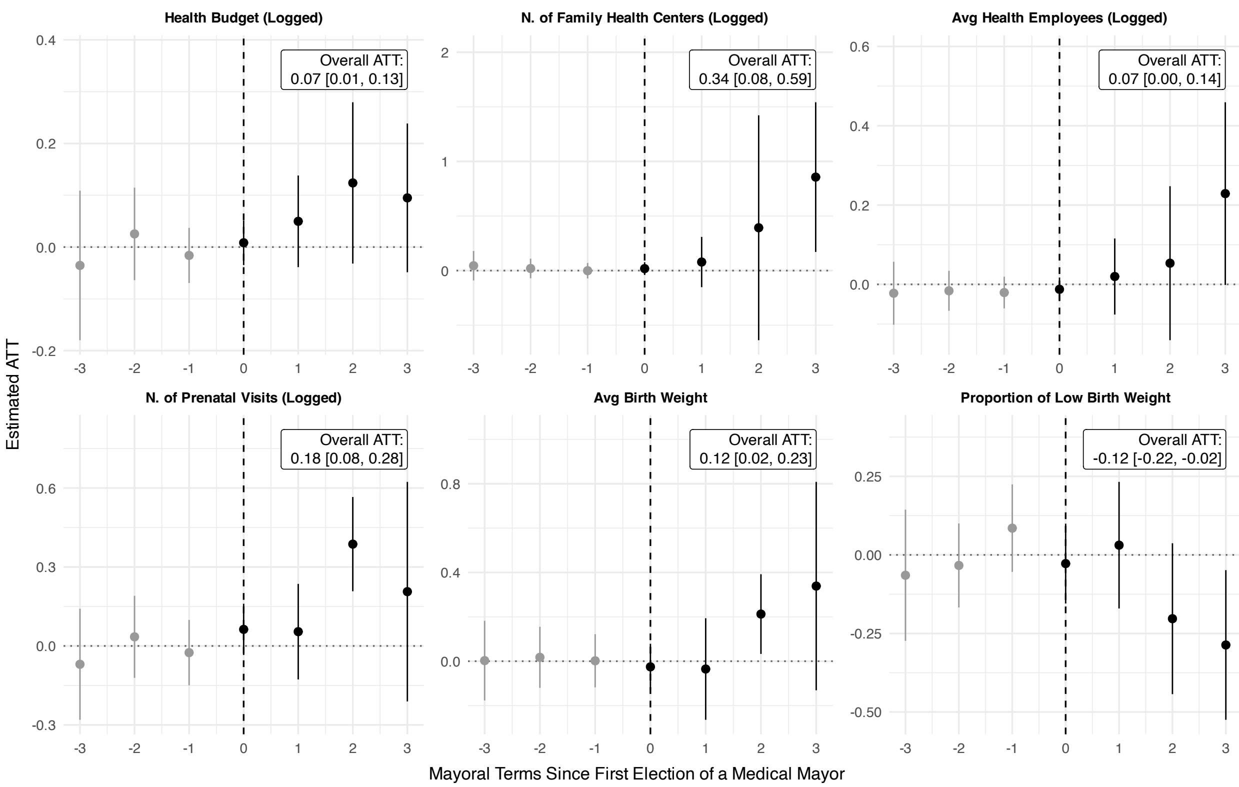


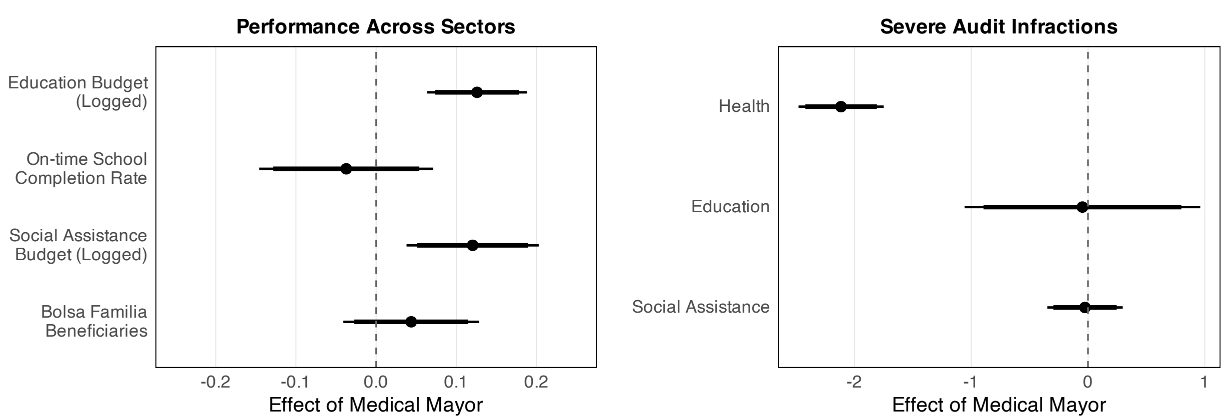
Figure Note: A value of -1 on the x-axis refers to the term immediately preceding the first term with a medical mayor in office, 0 to the first term in office, and 1 to the second term in office. The dataset includes 19,424 observations (municipality–mayoral terms). The x-axis represents the number of mayoral terms since electing a healthcare professional as mayor, while the y-axis represents the estimated treatment effect, expressed in standard deviation units. Vertical dashed lines indicate boundaries between successive four-year mayoral terms. Full results can be found in the supplementary appendix.

Our results highlight that expertise in government has the potential to generate deep institutional and organizational changes, but that these are unlikely to materialize quickly. Reforms in budgetary priorities, bureaucratic practices, and service delivery may require the gradual processes that unfold over years rather than within a single electoral cycle. This suggests that the potential of electing medical professionals as mayors is constrained by the electoral cycle itself: without continuity in office, the window for meaningful reform is too short. If meaningful change requires more than a single mayoral term, relying solely on the election of medical professionals at the municipal level proves insufficient as a public health strategy, and underscores the importance of designing health governance policies with longer time horizons than local electoral cycles typically allow.

Comparisons with Other Sectors: Limited Evidence of General Competence

In this section, we consider potential alternative explanations for the performance of medical mayors. First, we consider if medical mayors exhibit general – rather than sector-specific – competence. If this were to be true, we should see systematic improvements in performance in other sectors. We analyze spending and performance in education and social assistance, two areas where municipal governments in Brazil also hold substantial power. Specifically, we once again utilize a staggered difference-in-differences design to evaluate the education budget, the percentage of on-time school completion at municipal public schools, the social assistance budget, and the percentage of *Bolsa Família* beneficiaries, and provide the overall ATT. Results are provided in Figure 4 (left panel). While municipalities with medical mayors exhibit increased spending in both sectors, this increased investment does not translate into measurable improvements in sector performance, as we observed in the case of healthcare. This supports our theorized mechanism of sector-specific competence.

Figure 4: Medical Mayor Performance in Comparative Perspective



Note: Average post-treatment effects from staggered difference-in-difference estimation. Includes 90% and 95% confidence intervals. The left panel reflects performance of medical mayors in education and social assistance. The right panel reflects the discovery of severe infractions in health, education, and social assistance for randomly audited municipalities.

We also consider medical mayors' performance in random audits of financial mismanagement (e.g., embezzlement) in the healthcare, education, and social assistance sectors. This not only serves as a test of general vs. sector-specific competence, but also isolates whether medical mayors perhaps perform better in public health due to malfeasance or rent-seeking behavior. Results provided in Figure 4 (right panel) demonstrate that not only do medical mayors have a negative effect on malfeasance in the healthcare sector, but they have no effect on malfeasance in the areas of education or social assistance. This pattern suggests their outcomes are unlikely to be driven by rent-seeking behavior. Instead, it provides further evidence of sector-specific competence: medical mayors perform no differently in other sectors, but outperform their peers in their managerial quality in healthcare.

Additional Analyses & Robustness Checks

We conduct additional analyses that supplement our main findings. First, in Appendix B.5, we examine what happens when a municipality loses its medical mayor. To do so, we restrict the sample to treated municipalities and define the new treatment as the first electoral term after a medical mayor's term ends (and it is followed by a non-medical mayor). The new control group is composed of municipalities that retain their medical mayor. The logic is that if medical mayors are driving the improvements in health outcomes, their departure should decrease or stop the gains. This expectation is confirmed by the results. The overall ATTs following treatment removal provide non-significant negative estimates for most of them, which shows that the performance in health outcomes deteriorated or at least did not improve. These findings are consistent with the interpretation that our main estimates reflect the influence of medical leadership rather than changes that endure regardless of who governs.

We also engage in a series ofrobustness checks that demonstrate the consistency of our findings. We re-estimate our models using non-logged outcomes (Appendix B.3), without pre-treatment covariates, and with a not-yet-treated control group instead of never-treated municipalities. Across all three alternatives, the conclusions are unchanged. We also show that results hold when using alternative definitions of the treatment, including variations in how we identify medical candidates from ballot names and registered occupations (Appendix B.7). We replicate our main analysis using the Sun and Abraham (2021) estimator (Appendix B.6) and the fixed-effects counterfactual (FEct) estimator (Liu, Wang and Xu, 2024) as alternatives to Callaway and Sant'Anna, producing substantively identical estimates. Additional analyses include using municipality-years as the unit of analysis (see the appendix), models using treatment dose as the treatment (Appendix B.9), comparisons of effects across treatment cohorts (Appendix B.8), and corrections for multiple comparisons (Appendix B.2). Results are consistent across specifications. An exploration of heterogeneous treatment effects by the mayor's party, ideology, previous health expenditures, and population (Appendix B.11) shows little systematic difference in outcomes across these categories. We also conduct a close-elections regression discontinuity design (Appendix C), but this analysis is only well-suited to test for short-term effects. In line with our findings, the as-if-random election of a medical medical has little statistically significant short-term influence on our outcomes of interest.[37]

The Components of Sector-Specific Competence

In the previous section, we highlight the overall impact of electing a medical mayor and the targeted impacts on public health. We argue this is evidence of sector-specific competence, which can be understood as comprised of three components: 1) *issue priority*, 2) *technical knowledge*, and 3) *professional capital*. To investigate these components, we draw on an analysis of campaign documents and our bureaucratic survey.

To be sure, our evidence regarding medical mayors' increased spending on healthcare vis-à-vis their peers provides initial evidence of increased issue priority. But, we also consider a measure of issue priority before entering office, helping to guard against the possibility that issue attention may be shaped by their governing context, rather than an ex-ante individual characteristic. Our analysis of over 35,000 campaign platforms, including about 10% medical candidates, via seeded LDA models paints a clear picture. As shown in Figure 5, mayoral candidates with medical backgrounds emphasize healthcare more than their peers (0.17 SDs). While they also dedicate more attention to social policy, the magnitude of this effect is one quarter the size (0.043 SDs). In Figure B4, we consider one other measure of issue priority: funding

[37] Although we do find a statistically significant and positive effect for hiring health employees in this design.

requests to the federal government. We find that medical mayors request more healthcare-specific funding compared to their peers – including as early as the first treated period – a pattern not systematically detectable in other policy areas.

Figure 5: Issue emphasis in campaign platforms

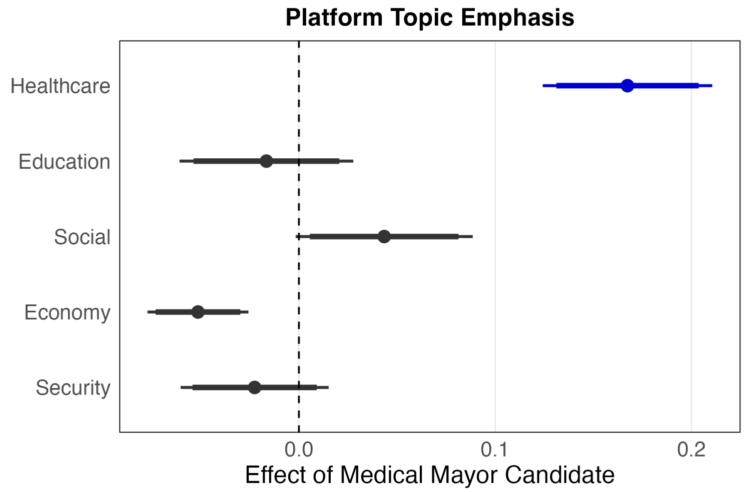


Figure Note: Outcomes are standardized proportions of platform emphasis on topics, as measured by a seededLDA models. Models incorporate municipality and year level fixed effects, and control for candidate gender, age, higher education attainment, and party affiliation, as well as municipality population, number of Bolsa Familia recipients per capita, and tuberculosis cases per capita. Standard errors are clustered at the municipality-election year level. Coefficients reported with 90% and 95% confidence intervals. N = 34,760.

Moreover, an analysis of how candidates discuss health in their platforms provides evidence of differences in *technical expertise*. Figure 6 plots the semantic space around the world "health" (*saúde*) drawing on context-specific word embeddings (Rodriguez, Spirling and Stewart, 2023). These results demonstrate that, on average, mayoral candidates with healthcare backgrounds discuss "health" differently than their non-medical peers. In particular, they emphasize family and primary care, as well as other specific forms of care (e.g., oral care, mental healthcare, psycho-social care). On the other hand, non-medical candidates tend to accompany discussions of health with more generic terms, such as "implementation," "secretariat," "population," and "patient." This provides evidence that medical professionals may be more knowledge about specific forms of healthcare, and are more likely to emphasize well-known levers to improve health outcomes (e.g., primary care). While ratios are small, they originate form a high-dimensional and dense embedding space, and focus on words already deeply in the same semantic region.[38]

Evidence from our bureaucratic survey supports both the *technical expertise* and *professional capi-*

[38]We have 9,935 word types in our vocabulary, with each word as a 300-dimension vector.

Figure 6: Differences in semantic space of discussions of "health" in campaign platforms

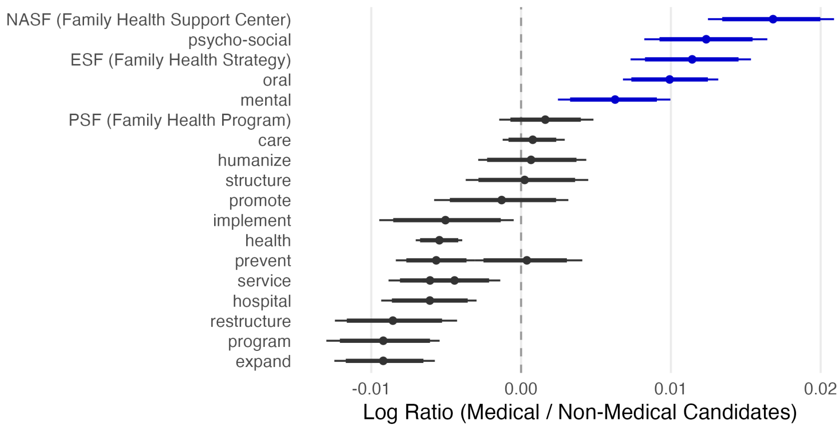


Figure Note: Plot represents the nearest-neighbor salience around 'saúde' (health). Point estimates correspond to log ratios with medical candidates in the numerator, non-medical candidates in the denominator. Results derived from the `conText` package and capture divergences in how words are used by different groups using word embeddings.

tal mechanisms, as well as our general conclusions. Results from our bureaucratic survey are presented in Figure 7. In general, responses from healthcare bureaucrats report overall better performance and approval for medical mayors. For example, bureaucrats working for medical mayors are more likely to report higher-quality service delivery (middle panel) and fewer problems with meeting resource demands (right column). But specific responses also lend credence to our proposed mechanisms. In particular, general higher levels of approval (Figure 7, left panel) indicate that healthcare bureaucrats tend to have more confidence in a medical mayor, an indicator of greater support and deference; this is directly related to *professional capital* (Kuo and Kelly, 2026; Costa and Pereira, 2024). Responses from bureaucrats also support the *technical knowledge* mechanism: respondents with medical mayors report on average that their mayors have specialized knowledge.

Qualitative evidence from open-ended survey responses support these mechanisms, as well. One respondent comments directly on their mayor's technical abilities, attributing it to their work in public health and on-the-job expertise:

> "Our mayor understands the health area a lot; indeed, in addition to having been in municipal management, he also was a nurse. This gives him deep knowledge not only about the reality and needs of the municipality, but also about public workers' demands, especially in the area of health."

On the other hand, one healthcare bureaucrat working for a non-medical mayor notes a lack of technical expertise for their executive:

Figure 7: Relationship between medical mayor status and healthcare bureaucrat responses.

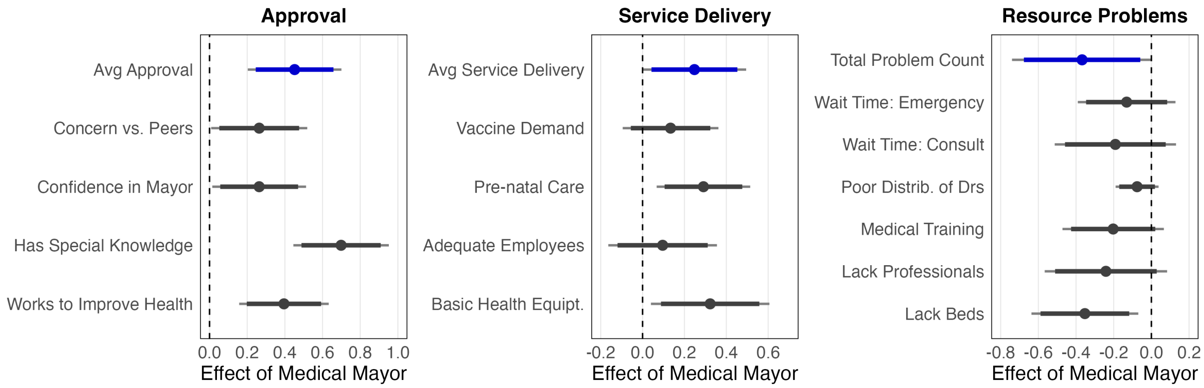


Figure Note: All responses are standardized. Models incorporate respondent-level weights and state-level fixed effects. Coefficients reported with 90% and 95% confidence intervals. Blue lines represent indices calculated as the average of responses in the relevant category (black lines).

> "He [the mayor] does not want to really understand how health services work, and goes about firing essential people and replacing them with incapable ones, all for political reasons."

Another bureaucrat working for a medical mayor expresses sentiments highlighting less specific technical expertise, but broader respect for their professional knowledge and deference to their judgment:

> "In addition to being a great healthcare professional, he is also a great human being and qualified to be where he is. He is able to put himself in others' shoes, he can listen to the population – and even more importantly – he really wants to resolve the problems that arise with dignified work and a huge heart. He deserves to be where he is, as a mayor, as a human being, and as a politician blessed by God."

Alternative Explanations

An additional alternative explanation for our results is that healthcare professionals may be more competent than their counterparts due to their above-average levels of education (Lahoti and Sahoo, 2020; Sørensen, 2023) or the fact that a sizable portion pursue political careers, leading them to have more government experience on average (Avellaneda and Gomes, 2017). Further, these factors could constitute a compound treatment or pose an issue of compensating differentials (Marshall, 2024). Consequently, improved public health outcomes may not result from their healthcare background but rather other common characteristics. To explore this, we study the effect of electing another type of highly educated mayor (lawyers) and the effect of electing mayors with previous political experience. If such politicians have similar impacts on healthcare as medical mayors, this would challenge our theory that sector-specific competence drives effects.

Results evaluating the election of a different highly-educated professional (lawyers) are shown in Appendix D.1. Using a similar staggered difference-in-differences approach, we find no evidence of a substantive nor significant effect on any of our outcomes. This pattern of null and noisy results stands in sharp contrast to the precise and directionally consistent effects we document for medical mayors, strongly supporting our contention that higher education alone does not translate into improved public health outcomes. Further, results evaluating the election of experienced politicians similarly show no pattern. In Appendix D.2, we provide results from a close-elections regression discontinuity design of individuals who have previously held political office.[39] Results show that there is no evidence of an effect of electing a previously experienced politician on institutional or community health outcomes. This supports the idea that political experience should not explain the positive impact of medical mayors.

Conclusion

This article studies the effects of expertise in public office by evaluating the political representation of medical professionals. We find that the election of medical professionals – such as doctors, nurses, and public health professionals – to mayoral office in Brazil results in tangible improvements to the public health field. Not only do these mayors invest more in the area, but they improve health outcomes (e.g., increasing average birthweight) and are less corrupt on average in this area. These conclusions are drawn from an analysis of over two decades of administrative data, leveraging a staggered difference-in-differences design. Our results are robust to multiple estimation strategies.

We argue that these effects can be attributed to *sector-specific competence*, comprised of technical knowledge, issue priority, and professional capital. Specifically, we argue that specialized professionals, like healthcare professionals, are able to deliver specific benefits to the sector they represent. We do not find evidence that medical mayors have broader benefits (e.g., in education or social assistance). We also rule

[39] We leverage an RDD, instead of a DID, because previous political experience among politicians is a very common factor, such that municipalities go in and out of treatment at high levels. Further, this leads to many "switchers," which results in a small sample size and undermines the quality of a staggered DID.

out prominent competing explanations: we find no evidence that the outcomes we identify are driven by mayoral competence – such as higher-than-average levels of education or political experience – or motivated interests.

Importantly, these findings should not be interpreted as a general argument for electing more highly educated or socioeconomically elite politicians. Research on representation shows that politicians from different economic backgrounds bring distinct priorities and perspectives to office (Carnes and Lupu, 2023), while formal education might not always be a reliable indicator of governing effectiveness (Carnes and Lupu, 2016). Instead, our argument is more specific: politicians whose professional backgrounds provide relevant technical knowledge, issue priority, and professional capital (i.e., sector-specific competence) may be better positioned to improve outcomes within their corresponding policy domain. As a result, expertise complements rather than displaces the broader democratic value of occupational and class diversity in public office.

In all, this paper makes a handful of contributions. First, we provide a systematic analysis – and demonstrate a clear impact – of the representation of public health professionals in public office. This contributes to the broader literatures on the politics of public health (e.g., Rich, 2013; Avelino, Barberia and Biderman, 2014; Niedzwiecki, 2014; Araújo, Gonçalves and Machado, 2017). Moreover, we make several contributions to the literature on occupational representation. First, we isolate a specific mechanism via which occupational representation may have impacts – sector-specific competence. Second, we provide a more hopeful analysis, as much of the literature has focused on the negative effects of such representation, and how certain occupational categories may benefit their in-group alone, often harming the masses (Szakonyi, 2018; Fuhrmann, 2020; Novaes, 2024). Instead, we demonstrate evidence of positive effects of occupational representation from a specific profession. Although these benefits are limited to their sector, we identify no risks to other analyzed policy areas.

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